

Dev Garg

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PROFESSIONAL SUMMARY

Software Engineer with ~4 years specializing in agentic AI systems and scalable backend infrastructure. Designs and ships multi-agent orchestration, RAG pipelines, and LLM-integrated tools, grounded in distributed-systems and event-driven architecture on AWS, owning systems from design through production.

EXPERIENCE

Encando.AI

Founding/Senior Software Engineer

College Station, TX · May 2025 - Present

- Architected a scalable, AI-first LMS backend from zero-to-one, serving 5,000+ users, with REST APIs and event-driven system architecture on AWS (ECS, DynamoDB, Redis) that reduced median latency from 1.4s to 67ms and improved P95 by 3.5×
- Designed and shipped a production agentic AI system - specialist sub-agent routing, custom long-term memory and compaction, multi-workflow orchestration, and a multi-stage RAG pipeline - serving 5,000+ users, built on LangChain, Pinecone, and OpenAI, with response relevance improved by 20% (Ragas).
- Enabled reliable releases by leading a team of 5, establishing CI/CD pipelines, code review standards, and agile workflows; decoupled long-running AI inference from user-facing APIs via AWS Batch with concurrency controls and failure handling - collaborated directly with customers to co-architect and ship multi-step AI workflows at scale.
- Established end-to-end ownership of the platform, analyzed and optimized system efficiency, reducing monthly AWS infrastructure costs by 15% and cutting AI inference costs by 35% through efficient prompt design.

Societe Generale GSC

Software Engineer

Bengaluru, India · May 2021 - Dec 2023

- Led the development of streaming data processing pipelines for credit risk analysis, enabling high-volume data ingestion for downstream analytics by utilizing Big Data technologies like Data Lake, Apache Spark and Kafka.
- Accelerated system validation cycle times by 50% and reduced quarterly compute costs by €20,000 by building an automated regression testing suite, designing auto-scaling data pipelines and building telemetry and observability dashboards using Grafana.
- Built and optimized distributed data workflows using Java, Spring Boot, SQL, and Scala, supporting critical analytics capabilities for real-time risk management applications on Dataplane and Kubernetes.

Software Engineer Intern

Remote · May 2020 - Jun 2020

- Delivered an NLP-powered incident resolution MVP in under 2 months, improving operational response times by 38%, using recommendation logic leveraging similarity-based retrieval over historical resolution logs, collaborating with cross-functional teams.

EDUCATION

Texas A&M University

Jan 2024 - May 2025

Master of Computer Science, Computer Science

- **GPA:** 4.0/4.0
- **Coursework:** Large Language Models, Software Engineering, Deep Learning, Info Storage and Retrieval

Indian Institute of Technology, BHU

2017 - 2021

Bachelor of Technology, Electronics Engineering

- **GPA:** 8.78/10.0

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, C++, SQL, TypeScript
- **Frameworks:** PyTorch, LangGraph, OpenAI API, Scikit-learn, LangChain, Spring Boot, Apache Spark, React
- **Platforms:** AWS, Kubernetes, Docker, Azure, GCP, ElasticSearch
- **System Design & AIOps:** System Architecture, Distributed Systems, Agile Methodologies, CI/CD, Claude Code, Vector Databases
- **Certifications:** Google Cloud Professional MLE, Machine learning Spec., MLOps Spec., GenAI with LLMs

PROJECTS

PaperFormer: A Citation-Graph Enhanced Language Model for Scientific Applications.

- Achieved State-of-the-Art ROUGE-1 scores (47.85) and reduced perplexity by 52% by architecting a Llama based model enhanced with novel inter-document attention mechanisms to process citation graphs.
- Outperformed GPT-4o in human-aligned review generation by LoRA Fine-tuning a quantized LLM model using PyTorch across multi-GPU HPC infrastructure, optimizing distributed training workloads for memory-constrained environments.
- Engineered context-window compression strategies to fit multi-document citation graphs into 512-token embeddings.

News Aggregation and Recommendation System

- Built an end-to-end MLOps pipeline using MLflow + Kubeflow for experiment tracking, drift detection, and automated retraining, maintaining model F1-scores above 0.82, while leveraging Kubernetes and Kafka to achieve a 40% reduction in system latency during stress testing.
- Leveraged PySpark's distributed ALS algorithm, reducing model training time by 90%, for a hybrid recommendation engine, achieving <50ms retrieval latency by implementing an offline pre-computation strategy using FIASS.